

ALISTAIR RODRIGUES

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SUMMARY

Builds production-shaped LLM systems: a LangGraph agent with human-in-the-loop safety controls, an automated eval pipeline that catches agent regressions before they ship, and a hybrid-search RAG pipeline with its own evaluation suite. Comes to AI engineering from a data analysis and UI/UX design background — comfortable across the data, the model, and the interface, not just one layer of the stack. Currently deepening this foundation through a structured AI engineering course, with RAG, agents, and evals as the core focus.

SKILLS

AI & LLM Engineering: LangGraph, Claude API & LLM integration, Retrieval-Augmented Generation, hybrid vector + keyword search, LLM-as-judge evaluation, agent regression testing, Ollama, Groq

Backend & Infrastructure: Python, FastAPI, PostgreSQL, Redis, DuckDB, Docker, Supabase, Git, CI/CD (GitHub Actions)

Data & Analytics: SQL, Pandas, NumPy, Scikit-learn, R, MATLAB, Matplotlib, Seaborn, Tableau, Power BI, Streamlit

Frontend & Product: React, HTML/CSS, Three.js, WebGL, Figma, UI/UX Design

Collaboration: Agile/Scrum delivery, stakeholder communication, team leadership

PROJECTS

agent-orchestrator — Multi-Step AI Agent | *Python · LangGraph · Claude API · FastAPI · Redis · PostgreSQL · 2026*

- Built a production-shaped agent (Plan → Act → Reflect) on LangGraph and the Claude API, with typed tools and a human-in-the-loop approval gate that blocks dangerous actions before they execute.
- Persisted session state in Redis and wrote every agent decision to a replayable PostgreSQL audit log, so any run can be reconstructed step-by-step after the fact.
- Exposed the agent through an async FastAPI service backed by 19 passing tests, with a no-API-key demo mode so reviewers can run it without provisioning credentials.

agent-evals — Agent Regression Testing | *Python · Claude-as-Judge · DuckDB · Streamlit · 2026*

- Built an automated eval pipeline that answers one concrete question: did the new agent version regress? Ingests real agent conversation logs from Postgres, clusters prompts with k-means (math, reports, RAG, safety...), and has an LLM judge generate a "golden" answer set with rubrics for the highest-value and highest-risk cases.
- Scores any new agent run against the golden set on correctness, groundedness, and safety (1–5 each), storing results in DuckDB and surfacing regressions through a Streamlit dashboard.
- Built to run against the same kind of replayable Postgres audit trail agent-orchestrator produces — closing the loop from "the agent made a decision" to "here's proof it's still making good ones."

ragstar — Hybrid Search RAG | *Python · Ollama · Vector + Keyword Search · 2026*

- Built a hybrid-search RAG pipeline merging vector (semantic) and keyword (exact-match) search, generating temperature-0, citation-grounded answers from a fully local LLM via Ollama — zero external API keys.
- Shipped an interactive web UI that visualises every retrieval stage end-to-end, alongside a CLI eval suite scoring retrieval quality and answer faithfulness against a 10-question benchmark.
- Designed for inspection over black-box output: every answer traces back to its source documents, and background model warm-up keeps the demo usable from first load.

PROFESSIONAL EXPERIENCE

UI/UX Intern — CMS Info Systems Ltd · *Mahape, Navi Mumbai, India* | *Mar 2026 – Sep 2026*

- Designing wireframes and interactive prototypes for internal IT tools, working directly with developers to keep interface decisions grounded in technical constraints and system behaviour.
- Running usability reviews on existing interfaces, using heuristic evaluation and flow mapping to turn friction points into a concrete list of fixes.

- Delivering design work inside Agile sprint cycles — joining stand-ups and reviews with the IT team from kickoff through release.
- Applying structured, evidence-based research methods to ground every design decision in data rather than opinion.

Events Team Leader — Edinburgh International Conference Centre (EICC) · *Edinburgh, UK* | *Jan 2024 – Dec 2024*

- Led a team of crew members across high-profile international conferences, coordinating task allocation, managing workflows, and keeping event objectives on schedule.
- Acted as the primary client-facing point of contact during live events, resolving escalations and holding service standards under pressure.
- Coordinated logistics across concurrent event streams, adapting plans in real time as schedules, attendance, or client requirements shifted.
- Ran a structured feedback loop on post-event data — analysing customer responses to surface recurring gaps and feed them back into the next event's plan.

Data Analysis Intern — Binex Controls · *Pune, India* | *Dec 2022 – Jul 2023*

- Cleaned and preprocessed datasets in Python (Pandas, NumPy), resolving missing values and inconsistencies to improve downstream reliability.
- Built visualisations in Matplotlib and Seaborn to surface trends across multiple datasets, supporting internal reporting and business decisions.
- Presented findings monthly to non-technical stakeholders, translating model outputs into clear, actionable summaries.
- Collaborated with the development team to feed analysis outputs into internal tooling, bridging data work and engineering delivery.

EDUCATION

MSc Advanced Software Engineering — University of Strathclyde · *Glasgow, United Kingdom* | *2023 – Present*

BSc Computer Science — Savitribai Phule Pune University · *Pune, India* | *2019 – 2022*

Relevant Coursework: Data Structures, Algorithms, Machine Learning, Statistical Analysis, Database Systems

AI Engineering Intensive (in progress): applied program covering RAG systems, agents, and FastAPI-based LLM services.

VOLUNTEERING

Data Volunteer — Umbrella Health Partnership · Birmingham City Council Youth Service · *Aug 2023 – Present*

- Built the organisation's data system from scratch, turning paper-based, unstructured records into a clean, structured, analysis-ready single source of truth.
- Used the new structure to run quarterly analysis, surfacing trends and gaps that fed directly into leadership's programme decisions.